



Language control and lexical access in diglossic speech production: Evidence from variety switching in speakers of Swiss German



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ABSTRACT

Diglossic speakers of two varieties of a language switch between their language varieties as bilinguals do between their languages, instead of having the otherwise typical probabilistic distributions of variants across varieties. To investigate the mechanisms involved in diglossic language control and lexical access, we conducted two switching experiments, in which participants who were highly proficient in the two varieties, or languages, named pictures in Swiss German or Standard German (Experiment 1), or in Swiss German or Tamil (Experiment 2), with pictures having cognate or non-cognate names in Swiss German and Standard German. The overall magnitude of switch costs in diglossic switching was comparable to switch costs in bilingual switching. Switch costs were asymmetric, favouring switching into the sociolinguistically preferred variety (Swiss German) in Experiment 1 and the slightly less dominant language (Tamil) in Experiment 2. Cognate gains were higher for Swiss German than Standard German and occurred even for Tamil responses. These results are compatible with a model that assumes that diglossic switching is influenced by similar mechanisms of language control as bilingual switching, that inhibitory processes are involved, that the amount of lexical inhibition depends on the relative strengths of lexical representations and on top-down control, and that diglossic cognates have integrated lexical representations.

Introduction

The remarkable ability of bilingual speakers to separate their languages, to control their language output, and to switch between languages voluntarily has attracted much attention in research on bilingual processing, including studies of acquisition courses (e.g., Core, Hoff, Rumiche, & Señor, 2013; Meisel, 1989; Paradis & Genesee, 1996), the potential effects on executive abilities (e.g., Bialystok, 2009; Bialystok, Craik, & Luk, 2012; Costa, Hernández, Costa-Faidella, & Sebastián-Gallés, 2009; Rubio-Fernandez, 2017), and the mechanisms involved in language control and lexical access (e.g., Christoffels, Firk, & Schiller, 2007; Costa & Santesteban, 2004; Green & Abutalebi, 2013; Meuter & Allport, 1999). However, the need to separate language systems, to control them and to restrict lexical access to items in one of their lexicons may also arise in speakers of a language with sharply distinct language varieties – as is the case in diglossia. In this study, we explore intentional language switching between varieties of a language in diglossic speakers, and the mechanisms and underlying representations involved in restricting lexical access to only one variety. We will do so

by comparing the patterns of diglossic and bilingual switching (in native speakers of Swiss German), studying the effects of switching and cognate word status onto response times.

Variety switching and diglossia – The case of German-speaking Switzerland

In German-speaking Switzerland, native speakers of German have command of two varieties of German, which are used in different situations: *Swiss German* (umbrella term for all of the Alemannic dialects spoken in Switzerland) and *Standard German* (the standardised and codified German, used in most writing and in educational contexts).

Such a juxtaposition of two language varieties in a society, such as Swiss German and Standard German in Switzerland, is called *diglossia* (e.g., cf. Ammon, 2003; Ferguson, 1959). It contrasts with both *bilingualism*, and smooth transitions between linguistic varieties in *dialect-standard continua*. While linguistic choices in bilingual speakers occur in terms of language choice or code-switching, linguistic choices associated with intra-language varieties are usually described as manipulations of linguistic variables, e.g., phonetic or morphosyntactic

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variables (Wolfram, 2007), with frequency of use depending on social and situational factors (cf. Labov, 2004). Indeed, it has been emphasised that “the distribution of variants is not held to be ‘either/or’, but rather probabilistic” (Watt, 2007, p. 5).

In contrast, in diglossic speakers linguistic choices occur in terms of variety choice and code-switching – as is the case with bilingual speakers. There is not a probabilistic, but a dichotomous distribution of linguistic variants across varieties (see Vorwerg, 2019). In each particular situation, one speaks or writes distinctly either Swiss German (SwG) or Standard German (StG). Accordingly, there is no continuum between Swiss German and Standard German (cf. Ammon, 2003) – neither along situations (i.e., no style-shifting within speakers) nor along social groups (i.e., dialect use by all social layers). To use the words of Ferguson (1959, p. 325), who introduced the concept of diglossia into sociolinguistic research taking Swiss-German as one defining example, the two varieties “exist side by side throughout the community”.

Historically, the German standard language developed out of a writing language, which spread, accompanied by continuous forming and equalising processes, from an East Middle German area to other German-speaking areas (see Besch, 2003). As such, it originated and developed far outside of the Swiss language areas, and its societal position relative to the Swiss German dialects has had a long history of debates and public discourse (see Sonderegger, 2003). Nowadays, SwG and StG are recognised as two varieties that exist side by side, each in their own right, the choice between which depends on function and situation. Linguistically, they differ in characteristic ways across all domains (see, e.g., Christen, Glaser, & Friedli, 2010; Rash, 1998), such as phonetics/phonology, morphology, lexis, and syntax (see Appendix A for a detailed description of the linguistic situation). For oral communication, SwG is generally preferred, while StG is used for most written texts. By most SwG speakers, StG is only used spoken if there is a particular reason, such as a rule in educational contexts or the presence of people who do not understand SwG. Speakers may alternate between both varieties, if the situation changes in those regards (e.g., between class and more private group discussions), or when quoting written language in an otherwise SwG conversation; however, code-switching (switching within a speech event) is rare and speakers try to avoid mixing varieties. Given that SwG/StG diglossic speakers restrict lexical access to one language variety or the other (similar to bilingual speakers), we investigated what mechanisms are involved in this and whether they are similar to those controlling lexical access in bilingual speakers, using the paradigm of cued switching.

We also studied bilingual switching between one of the two German varieties (SwG) and a second language (Tamil) in bilingual speakers, for the sake of a direct comparison between diglossic and bilingual switching. Children with an immigrant background growing up in the German part of Switzerland, whose parents speak a different language, usually grow up with two languages in parallel, and this is also true for the Swiss Tamils. They use their two languages in different contexts, and they must learn the German language in two varieties: SwG for child-peer communication, and StG to attend school. So, they usually speak StG in class, SwG with their friends (including more relaxed school situations and breaks), and their parents’ language at home (see, e.g., Ender & Straßl, 2010). Accordingly, the young Tamils in Switzerland have been described as having “different lives inside and outside the home” (Swissinfo, 2010) – an assertion that certainly goes beyond language use, but includes it as well. The SwG/Tamil bilingual speakers are therefore comparable to the SwG/StG speakers in terms of using their respective varieties predominantly in different contexts.

Lexical access mechanisms and switching effects

Both monolingual and bilingual models of lexical access in speech

production usually assume at least three levels of representation – the conceptual, the lexical and the phonological one – and two stages of processing: lexical selection and phonological encoding. The oral production of a word starts with the activation of a lexical concept (i.e., a concept that is linked with a verbal label), which spreads activation both to other related concepts in the conceptual network and to its corresponding lemma (abstract lexical representation of a word that specifies its syntactic properties) in the lexical level. The first stage of lexical access is lexical selection: the retrieval of a lemma from the mental lexicon via selecting it from a set of activated candidates (e.g., Levelt, Roelofs, & Meyer, 1999; for a discussion of selection mechanisms proposed, see La Heij, 2005). The second stage encompasses the retrieval of the word form, i.e., the word’s phonological shape, from the mental lexicon, involving activation spreading from the lemma (or lexical) level to the phonological (or sublexical) level.

The need for language control arises in models of bilingual access, which usually assume that almost every lexical concept in the bilingual lexicon is linked to two lexical representations (Costa, 2005), or at least that concepts from both languages share properties (Green, 1998). There is also wide agreement that activation from the conceptual system flows to both lexicons, i.e., is language-unspecific (see Costa, 2005, for a review). Two main proposals have been put forward for the mechanism ensuring that a linguistic item from the intended language is selected: reactive inhibition and language-specific selection. Both theoretical approaches draw upon switching effects found with the cued switching paradigm. In this paradigm, speakers name pictures of objects or digits in one language or the other depending on a cue presented, such as a color cue. Naming latencies have consistently been found to be larger in switch trials (different language than in the trial before) than in non-switch trials (same language as in the trial before) (e.g., Christoffels et al., 2007; Costa & Santesteban, 2004; Meuter & Allport, 1999). The switching costs may or may not be asymmetrical across languages. A switch-cost asymmetry disfavoring the switching into the L1 or dominant language is described as “paradoxical”, as it takes more time to switch into the stronger language. The different response time patterns found for different linguistic constellations are thought to reveal different mechanisms of language control.

One theoretical approach, the IC (inhibitory control) model developed by Green (1998) assumes that the lexical items of the non-target language are reactively inhibited after being activated, preventing their selection. This mechanism, which takes place at the lexical level, presupposes that the lemmas are flagged with a language tag, and that there is a mental device which exerts inhibitory control: language task schemas, which, in turn, are modulated by a higher level of control (Green, 1998). According to the IC model, switch costs stem from (1) the change of the dominant language schema, and (2) overcoming the inhibition of the previous language tags. Most importantly, the IC model predicts asymmetrical switch costs for switching into a language that was more strongly suppressed (Green, 1998). For unbalanced bilinguals, this is assumed to be their dominant language (Green, 1998), which is harder to suppress than the weaker language, as evidenced by greater within-language interference and smaller between-language interference in Stroop tasks (Mägiste, 1985). A number of studies yielded asymmetrical switch costs for unbalanced bilinguals in the direction predicted by the IC model (Costa & Santesteban, 2004; Meuter & Allport, 1999; Macizo, Bajo, & Paolieri, 2012; Peeters, Runnqvist, Bertrand, & Grainger, 2014; Reynolds, Schlöfel, & Peressotti, 2015; for a review, see Reynolds et al., 2015). Other studies have presented evidence suggesting that the switch-cost asymmetry cannot be reduced to an effect of response selection as proposed by Finkbeiner, Almeida, Janssen, and Caramazza (2006) or other language-external processes, and that it arises at least in part from processing in the speech production system (Peeters et al., 2014; Reynolds et al., 2015).

An alternative theoretical approach proposes that in highly

proficient bilinguals language selection is language-specific, i.e. restricted to the intended language (Costa & Santesteban, 2004; Costa, Santesteban, & Ivanova, 2006; Martin et al., 2013; for a review see Reynolds et al., 2015²). According to this view, the language selection mechanism does not consider activated lexical items that do not belong to the intended language, i.e., lexical selection would proceed as if the lexical items of the alternative language were not activated (cf. Costa, 2005). Evidence in favour of this account are symmetrical switch costs (instead of an asymmetry) in highly proficient bilinguals, not only when switching between their two dominant languages, but even when switching between L1 and a much weaker L3 (Costa & Santesteban, 2004; see also Martin et al., 2013). This account relegates the mechanism of reactive inhibition proposed in the IC model to unbalanced bilinguals, while acquiring proficiency in a second language also entails acquiring a new lexical selection mechanism, one that is language-specific (Costa & Santesteban, 2004). Once acquired, this new selection mechanism would then also be effective with other, weaker languages spoken by the same individual. However, asymmetric switch costs were observed in the same individuals, when one of the languages involved was a very weak language. This suggests that there are limits to the functionality of the restricted-selection mechanism and even highly proficient bilinguals need to resort to inhibitory control in certain conditions (Costa et al., 2006).

The pattern of findings becomes, however, more complicated by two other observations. First, symmetrical switch costs can occur even with moderately proficient bilinguals (Christoffels et al., 2007). The authors attributed this finding to the fact that the bilinguals involved switched between languages continuously in their daily lives. This argument is in line with the proposal that language control might depend on the way that bilingual speakers typically use their two languages, in particular whether they are frequently in a dual language context (such as in code-switching communities) or frequently in a single language context (such as in non-code-switching communities) (Green, 2011).

Second, there is frequently a general response time disadvantage of the stronger language in switching contexts (i.e., slower response, whether switch or not), which also seems “paradoxical”. This effect occurred in all experiments involving highly proficient bilinguals reviewed above, but also in two other studies yielding symmetrical switch costs, which were attributed to either a high level of switching experience in daily life (Christoffels et al., 2007) or long preparation intervals (Verhoef, Roelofs, & Chwilla, 2009). The effect was also observed in some (but not all) of the experiments that found asymmetrical switch costs. The L1 speed disadvantage might stem from a general inhibition or higher selection thresholds for the lexical items of L1 across the whole set of trials (cf. Costa & Santesteban, 2004; Costa et al., 2006). It is, however, difficult to see why such a mechanism should be necessary if lexical selection was language-specific.

Taken together, naming-switching studies have consistently found at least one of the paradoxical effects: either a switch-cost asymmetry or a language effect disfavoured the stronger language. We conclude from the review of the literature presented that the most parsimonious account of the data pattern is language-unspecific lexical selection with the need to inhibit items of the unintended language and two possible mechanisms to counteract an imbalance in activation levels: global inhibition (or threshold-level increase), or reactive inhibition. Accordingly, global language effects and switch-cost asymmetries should be considered in relation to each other.

In contrast to the bilingual models discussed above, monolingual models of language production do not pose the question of how to restrict lexical competition, and they usually assume that each lemma is

uniquely linked to a lexical concept (e.g., Levelt et al., 1999; see above). However, given that diglossic speakers in German-speaking Switzerland have to restrict lexical selection to the lexical representations in one language variety, they should need to control their varieties in a similar way that bilingual speakers control their languages. If this is the case, we would expect similar patterns of switch costs for switching between Swiss German (SwG) and Standard German (StG) as in bilingual language switching.

According to the IC model, language switching takes time because it involves inhibiting the previously active language schema and overcoming the inhibition of the previously irrelevant language. Thus, if diglossic speech control involves language task schemas, such as ‘word production in SwG’ and ‘word production in StG’, the overall amount of switch costs in diglossic switching should be comparable to bilingual switching. In addition, for both diglossic and bilingual switching the question is how individuals who use two strong languages or varieties predominantly in single language context, i.e., without much code-switching, manage to restrict lexical access to one of them. Switch-cost asymmetries would be indicative of a stronger degree of suppression for one of the varieties or languages (effective when being the unemployed alternative). A global language effect (latency advantage for one of the two varieties or languages) would be indicative of its activation increase (effective when being the required variety or language).

Lexical access mechanisms and cognate effects

Another effect found with the language switching paradigm is the cognate facilitation effect: the finding that naming latencies are shorter for pictures whose names are cognates in the two languages (translation equivalents with similar phonological forms) than for pictures whose names are non-cognates (Christoffels et al., 2007; Costa, Caramazza, & Sebastián-Gallés, 2000; Verhoef et al., 2009; for reviews, see Costa, 2005; Roelofs & Verhoef, 2006). The cognate facilitation effect is usually assumed to arise during postlexical processing, e.g., phonological encoding (e.g., Christoffels et al., 2007; Costa et al., 2000; Roelofs & Verhoef, 2006; Sadat, Martin, Magnuson, Alario, & Costa, 2016; for discussions of alternative accounts, see Costa, Santesteban, & Caño, 2005; Hanulová, Davidson, & Indefrey, 2011). It has accordingly been regarded as evidence for the assumption that the concurrent activation flow to representations of both the intended and the alternative language extends to the sublexical level in a cascading manner (Costa, 2005; Roelofs & Verhoef, 2006; for a related discussion, see Costa et al., 2005). According to this account, the phonological activation of a cognate translation equivalent facilitates picture naming because it shares phonemes with the target and its activation sends additional activation to the target’s phonemes. Additionally, there may be feedback from the translation equivalent’s shared phonemes to the lexical level (Berndt, Hartsuiker, & Pickering, 2012).

For bilingual representations, the idea that the cognate facilitation effect arises at the postlexical stage is consistent with evidence from syntactic priming suggesting that even in closely related languages, such as Mandarin and Cantonese, cognates have separate rather than shared lexical representations across languages (Cai, Pickering, Yan, & Branigan, 2011). However, for the diglossic representations of SwG and StG, experimental evidence gained through the same priming paradigm supports the view that across varieties of one language, cognates may share lexical representations (Lüthi, Vorweg, Pickering, & Branigan, 2019b).

In the present study, we explore SwG/StG cognate effects in both diglossic switching between SwG and StG, and bilingual switching between SwG and Tamil. Cognate effects may indicate how much a non-target language is activated. We predict a cognate facilitation for SwG/StG switching. Based on the finding that cognate facilitation is larger in L1 than in L2 (Christoffels et al., 2007), it may be expected that there is larger cognate facilitation in SwG than in StG, inasmuch as SwG is the stronger oral variety. For the bilingual switching between SwG and

² Note two relevant errors in their overview table: Other than indicated there, the Martin et al. study observed symmetrical switch costs for the highly proficient L1-L2 group; furthermore, their L1-L3 group with asymmetrical switch costs comprised unbalanced bilinguals.

Tamil, cognate effects would indicate that StG is also activated, even though it was not used during the experiment. In case there are cognate effects in this situation, they can be expected to be positive (facilitation) in the SwG trials, and positive or negative in the Tamil trials, thereby serving as an index of whether cognate effects between SwG and StG occur at the phonological or at the lexical level.

Experimental outline

The main goal of the present study was to investigate the effects of language or variety switching and cognate status between varieties on word production in (highly proficient) diglossic speakers. In the experiments reported below, we tested switch costs, cognate gains (or costs), and language effects in picture naming in speakers of Swiss German, taking the example of Bernese German.

In Experiment 1, speakers switched between two varieties of German, Swiss German (SwG) and Standard German (StG). In Experiment 2, speakers switched between one variety of German (SwG) and an unrelated language, Tamil (Tam). All speakers were highly proficient in both varieties of German, and the Tamil/German bilinguals also in Tamil. All participants had grown up in German-speaking Switzerland and used the relevant varieties (Experiment 1: SwG & StG, Experiment 2: SwG, StG & Tam) during childhood. Both groups of speakers use those varieties in different (single language) contexts; they do belong to non-code-switching communities. The speakers in Experiment 1 use SwG at home and with friends and in all more or less relaxed situations; they use StG in educational contexts (such as class and lectures) plus in the written modality (often except short exchanges in conceptually oral media, such as instant messaging and social media). The speakers in Experiment 2 use Tamil at home, SwG with friends, and StG in educational contexts.

If language control operates on similar mechanisms in diglossic as in bilingual speakers, switch costs comparable to available results from bilingual studies would be expected for cross-varietal switching. Considered together, Experiment 1 and Experiment 2 allow a direct comparison of diglossic and bilingual switching. Both experiments involve the same methodology and the same pictures; moreover, both include SwG as one variety and are performed with speakers who grew up in German-speaking Switzerland. According to the IC model, the overall amount of switch costs stems from the costs of (1) resetting the cognitive system and activating the new language schema, and (2) overcoming the prior inhibition of lexical items of the previously irrelevant variety. Both components can be expected to affect diglossic switching to the same degree as bilingual switching.

The second question addressed concerns language effects. Differences between the two languages or varieties involved may occur in the form of a switch-cost asymmetry or a general language effect. Our participants are highly proficient in both (or all three) varieties involved in the experiment. If a high level of bilingual proficiency leads to an absence of asymmetrical switch costs due to a different mechanism of restricting lexical selection, i.e. language-specific selection (as suggested by Costa et al., 2006; Costa & Santesteban, 2004), symmetrical switch costs should also be expected for SwG/StG switching – and also for SwG/Tam switching. Inasmuch as one of the varieties is stronger, a paradoxical language effect (slowing down of latencies of the stronger variety) should occur as well. If the absence of asymmetrical switch costs is, however, rather the result of a high level of switching experience (consistent with Christoffels et al., 2007), a switch-cost asymmetry would be expected, as both of our participant groups use their two languages/varieties predominantly in different contexts. In the SwG/StG community, code-switching is even rather frowned upon and the idea to keep varieties strictly apart is still alive in public discourse.

The linguistic situation in the German-speaking part of Switzerland is peculiar, in that the local dialects are highly valued societally. Rather than being at all regarded as quaint and old-fashioned or socially less appropriate than the standard language (as non-standard varieties are

in many parts of the world), they are understood to be the normal way of speaking. StG, on the other hand, is strongly present (e.g. in movies, many national TV programs, written materials, educational contexts, intercultural communication), but not preferred for spoken communication. SwG is regarded as one's own language and speakers might try particularly hard to avoid the intrusion of StG lexical items when speaking SwG. This might lead to a stronger suppression of StG than SwG (as proposed by Lüthi, Vorwerg, & Pickering, 2019a), resulting in asymmetrical switch costs in favour of the dialect, even though it is the first and orally possibly slightly more dominant variety. Following the IC model, it should take longer to switch into a variety that was previously more suppressed. While the minimum amount of inhibition required to suppress a dominant language in order to prevent lexical intrusions may usually be higher because its lexical representations are stronger, people may also feel a stronger need to suppress a language for motivational reasons. After all, there may be a variety of reasons why people try harder to avoid intrusions from an alternative language or task.

The third aspect investigated are cognate effects. Cognate facilitation effects (i.e., faster response latencies for cognates than for non-cognates) have been taken to indicate a concurrent activation of the non-response language's phonology (e.g., Christoffels et al., 2007; Costa, 2005; Costa et al., 2005). However, inasmuch as lexical representations are shared between varieties in diglossic speakers (as evidenced by Lüthi et al. results, 2019b), cognate gains in SwG/StG naming (Experiment 1) may also stem from the lexical level. For SwG/StG naming, the cued switching paradigm does not allow to distinguish between both possibilities. We predict cognate facilitation for SwG and StG, signalling that the alternative, task-irrelevant variety is active as well. However, for SwG/Tam naming (Experiment 2), different predictions result from the two assumptions about the locus of cognate effects. There are no direct cognates between Swiss German and Tamil (and all cognates resulting from English borrowings were excluded from the experimental materials in the cognate condition). Accordingly, if StG is activated despite not being overtly used, Experiment 2 serves for assessing the effects of “silent” (SwG/StG) cognates. If cognate effects stem (only) from phonology, a facilitatory effect would be expected for SwG responses, but an inhibitory effect for Tam responses. This is because phonemes shared between StG and SwG can only give additional activation to the SwG alternative making it more difficult to retrieve the Tam target word. If cognate effects stem, however, (also) from the lexical level, facilitatory effects would be expected for both languages, as even for Tam responses, less inhibition will be required if there are fewer activated non-target lemmas.

Experiment 1

Method

Participants

Seventeen SwG speakers living in the canton of Bern participated, all native speakers of the local Bernese German variety, a variety of SwG. Their mean age was 22.8 years (range from 18 to 28). All but one had grown up in the canton of Bern (one in the cantons of Fribourg and Bern). All participants reported speaking SwG with their parents and their siblings (one in addition Dutch). In addition to Swiss German (SwG) and Standard German (StG), participants knew two to five further languages. All reported knowledge of English and French, some of Spanish ($n = 6$) and/or Italian ($n = 5$). Other languages were mentioned by single persons. Participants' level of language proficiency was assessed for all languages and varieties using a self-rating proficiency questionnaire (adapted from Christoffels et al., 2007). They rated their proficiency in four areas of language skills (reading, writing, verbal expression, comprehension) and their general command of the language (variety) on a 7-point-scale (1 = very bad, 7 = very good). The mean scores for SwG and StG can be found in Table 1. The proficiency

Table 1
Proficiency questionnaire (SwG & StG) in Experiment 1.

	Swiss German		Standard German	
	Mean	SD	Mean	SD
Comprehension	6.8	0.6	6.7	0.6
Oral expression	6.6	0.7	6.2	0.9
Reading	6.5	0.7	6.6	0.9
Writing	6.2	1.1	6.0	0.9
General proficiency	6.2	1.7	6.2	0.7

profiles are remarkably similar and show a very high proficiency in both language varieties (general proficiency: 6.2 each). Only in oral expression, participants rated themselves slightly, but significantly [$t(16) = 2.13, p = .049$], better in SwG (6.6) than in StG (6.2). Accordingly, participants were considered to be highly proficient in both varieties, with SwG being their orally more dominant variety. When asked about their preference between both varieties, 15 participants indicated SwG and two StG. The first and second other languages (ordered by general proficiency or, if that was identical, by mean domain proficiency) had mean general proficiency levels of 5.4 and 4.4, and mean oral expression ratings of 5.0 and 3.6.

Materials

Twenty-four pictures of common objects were selected, which consisted of simple black-on-white line drawings. The objects were chosen such that they were easy to depict and name in SwG, StG, and Tamil (Tam). Accordingly, only concrete words were used.

Half of the pictures had cognate names in SwG (Bernese variety) and StG, the other half non-cognate names. The materials were constructed such that no picture in the SwG/StG cognate condition had a Tam cognate name. Generally, cognates between German and Tamil can only be borrowed cognates, mainly via English loanwords into Tamil. Three of the pictures in the SwG/StG non-cognate condition had Tam/StG cognates (*badder/Butter*, *motorcycle/Motorrad*, *carrot/Karotte*). Some cognate overlap of experimental words with words from other Indo-European languages is, of course, unavoidable. A comparison of names in the experimental varieties and the other languages spoken by more than two participants (English, French, Italian, Spanish) revealed that in the SwG/StG non-cognate condition 6 pictures (out of 14) had other-language names that are cognates to the StG names (e.g., *Kamm* / En. *comb*, *Bonbon* / Fr. *bonbon*, *Kaninchen* / Sp. *conejo* / It. *coniglio*), 2 had rather ‘hidden’ cognate relations to SwG (*Ross* / En. *horse*, *Stäbe* / En. *stairs*). In the SwG/StG cognate condition, there were 7 pictures (out of 14) with cognates to StG and SwG (all of them concerning English, e.g., StG. *Kuh* / SwG *Chue* / En *cow*, 2 in addition all other languages: the words for ‘eye’ and ‘ball’) – even though not all of them might be mentally represented as cognates (‘hidden’ cognates).

The selection of word triplets (SwG, StG, Tam) on the basis of the above-mentioned criteria did not allow for a perfect matching with respect to word frequency and word length. We report only the frequency of StG words (logarithmic frequency classes, based on full word forms, Quasthoff, Fiedler, & Hallsteinsdóttir, 2011), as for SwG frequency norms are not available. The word-length information refers to SwG and StG words. Cognate words tended to be of higher frequency than non-cognate words [mean: cognate = 10.0, non-cognate = 12.7; $t(22) = 3.6, p = .002$] and shorter, in terms of syllables [mean: cognate = 1.4, non-cognate = 2.0; $F(1,44) = 11.8, p = .001$], with word length being unaffected by variety or an interaction of variety and cognate status ($F_s < 1$). However, word length does not affect speech onset latencies when producing single words (Damian, Bowers, Stadthagen-Gonzalez, & Spalek, 2010). For word frequency, we controlled at the analytical stage, a posteriori.

See Appendix B for the full list of stimuli.

Design

Three factors were manipulated in this experiment: Language Variety (SwG or StG), Type of Trial (switch or non-switch), and Cognate Status (cognate or non-cognate). Participants were asked to name pictures (24) in either Swiss German or Standard German. The language variety was cued by the colour of a frame around the picture (blue or orange). The assignment of colour cue to language variety was counterbalanced across participants. A switch trial required a response in the other language variety than the one immediately preceding, while a non-switch trial required a response in the same variety. The order of trials was manipulated such that two switch trials and four non-switch trials occurred for each picture in each language variety ($24 \times 6 \times 2 = 288$ items, half of which with cognate names and half of which with non-cognate names).

Procedure

Participants were tested individually. Instructions were given on the screen in Standard German. Participants were instructed to name the pictures in either Swiss German or Standard German. They were informed that the language variety in which a particular picture had to be named was indicated by the colour of the frame around the picture. However, before the actual experiment, participants were familiarised with the pictures and their names in both varieties. During familiarisation, each picture was presented on the screen without a frame, with both designated names, and participants were asked to read them aloud. After familiarisation, there was a training block, in which each picture was presented twice, once with the orange frame and once with the blue frame, and had to be named accordingly, once in Swiss German and once in Standard German (48 trials). This was followed by six randomly presented experimental blocks of 49 trials each: 1 starting trial, 16 switch trials and 32 non-switch trials. Switching occurred unpredictably, with 0–3 non-switch trials intervening between switch trials. In each trial, the picture appeared with its coloured frame and remained on the screen for 2000 ms or until the participant responded. Then there was a blank interval of 1150 ms, before the next trial started. No feedback was given. The answers were audio recorded for later offline assessment.

Data analysis

At picture onset, a voice key was activated to measure naming latencies. The accuracy of latency measurements was checked with the help of the speech analysis software Praat (Boersma & Weenik, 2015). The following types of responses were scored as errors: (a) verbal disfluencies (stuttering, hesitation signals, utterance repairs, non-verbal vocalisations); (b) recording failures; (c) production of a definite or indefinite article preceding the word (‘article before noun’); (d) production of a different name from that designated by the experimenter (‘other name’); (e) naming in another language variety or language (‘other language/variety’). Deviations in word form, however, were not considered as different names (e.g., phonological variations, or the diminutive form *Bliemli* of SwG *Blume*), as long as the deviating form would not belong to the alternative language variety. Responses considered as errors were discarded from the latency analysis. Occasionally, an expected and correctly produced name occurred among a set of variable responses to a particular picture produced by a participant. Such individually correct, but inconsistent responses were also excluded from the latency analysis. Furthermore, naming latencies greater than 2000 ms or exceeding 2 SD from a given participant’s mean, calculated from all non-excluded trials, were discarded as outliers.

Results

Response analysis

Following the criteria above, 5.6% of the trials were determined as errors and 0.3% as individually correct, but inconsistent responses. Most errors were of the types ‘other language variety’ (2.0%) or

‘different name’ (e.g., SwG *Haas* ‘hare’ instead of *Chiingu* ‘rabbit’) (1.6%). A small number of the former (0.4%) were actually interference responses, with StG lexemes being phonetically adapted to SwG (e.g., the StG word *Abfall* ‘waste’ produced as *Abfau*, with Bernese *l*-vocalisation).

Trials with recording failures (0.5%) were excluded from further analyses. An error analysis for word errors (‘other word’, ‘other variety’, or ‘article before noun’) using a GLMM (generalised linear mixed model) with Language, Switch, and Cognate as fixed factors and Participant and Picture as crossed random factors (intercepts and slopes) and binary logistic regression as link function, implemented using the GENLINMIXED procedure in SPSS 24.0.0.1 (estimation with linearisation-based methods), revealed a significant main effect of Cognate [$F(1, 4865) = 8.26, p = .004$] and a significant interaction of Cognate and Language [$F(1, 4865) = 4.76, p = .029$]. Fewer errors occurred for cognate trials ($\hat{\mu} = 1.0\%$) compared to non-cognate trials ($\hat{\mu} = 3.4\%$). The difference was more pronounced for Swiss German ($\hat{\mu}_c = 1.0\%$ vs. $\hat{\mu}_{\text{non-c}} = 5.1\%$) than for Standard German ($\hat{\mu}_c = 1.1\%$ vs. $\hat{\mu}_{\text{non-c}} = 2.3\%$). No other main effect or interaction was significant.

RT analysis

Errors and inconsistent (correct) responses (see above) as well as outliers (4.8%) were excluded from the response time analysis. The effects of Language Variety, Switching, and Cognate Status onto response were analysed using a GLMM with Gamma distribution and Log link, with Participant and Picture as crossed random factors (intercepts and slopes), implemented using the GENLINMIXED procedure in SPSS 24.0.0.1 (estimation with linearisation-based methods). Table 2 lists the mean response latencies as a function of fixed-factor combinations.

There were significant main effects of Switching [$F(1, 4373) = 118.78, p < .001$] and Cognate Status [$F(1, 4373) = 33.59, p < .001$], indicating that switch trials were slower than non-switch trials (see Fig. 1), and that cognates were produced faster than non-cognates (see Fig. 2). Both effects were qualified by interaction with Language Variety. The significant interaction of Switching and Language Variety [$F(1, 4373) = 4.89, p = .027$] shows slightly asymmetrical switch costs, with SwG $\rightarrow$ StG $>$ StG $\rightarrow$ SwG (see Fig. 1). The also significant interaction of Cognate Status and Language Variety [$F(1, 4373) = 4.90, p = .027$] implies a slightly higher cognate effect for Swiss German (see Fig. 2). Simple contrasts confirmed for both varieties clear, statistically reliable cognate gains [SwG: 127 ms (SE = 36 ms), $t(4373) = -3.56, p < .001$; StG: 103 ms (SE = 31 ms), $t(4373) = -3.31, p = .001$] and switch costs [SwG: 45 ms (SE = 13 ms), $t(4373) = -3.54, p < .001$; StG: 67 ms (SE = 17 ms), $t(4373) = -3.96, p < .001$], which only differ in size. The cognate facilitation effect was significant for all type-of-trial and language-variety combinations [no-switch/SwG: 133 ms (SE = 37 ms), $t(4373) = -3.63, p < .001$; no-switch/StG: 100 ms (SE = 30 ms), $t(4373) = -3.29, p = .001$; switch/SwG: 120 ms (SE = 36 ms), $t(4373) = -3.33, p = .001$; switch/StG: 107 ms (SE = 34 ms), $t(4373) = -3.17, p = .002$]. Profile analyses plotting response-time data with word frequency on the x-axis confirmed a higher level for the non-cognate than the cognate profiles for both varieties, suggesting that

Table 2

Estimates for Mean Response Latencies in ms (with Standard Errors) [and 95% Confidence Intervals] as a Function of Language Variety, Switching, and Cognate Status for Experiment 1.

	Swiss German		Standard German	
	Non-Cognate	Cognate	Non-Cognate	Cognate
Non-switch	921 (211) [588, 1442]	788 (180) [503, 1234]	887 (203) [567, 1389]	788 (180) [503, 1233]
Switch	958 (219) [612, 1501]	838 (192) [535, 1312]	958 (219) [612, 1500]	851 (195) [543, 1333]

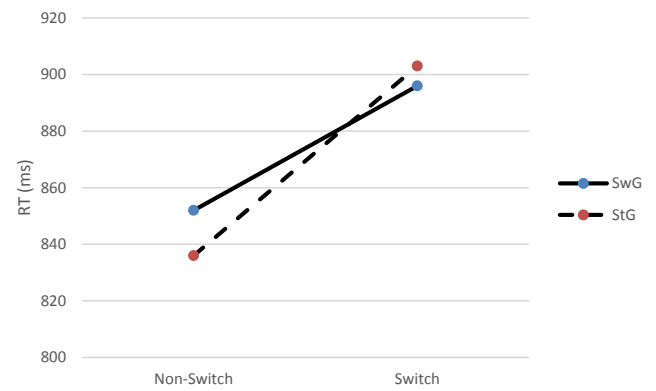


Fig. 1. Naming latencies (estimated means) for switch and non-switch trials for both language varieties in Experiment 1 (Swiss German and Standard German).

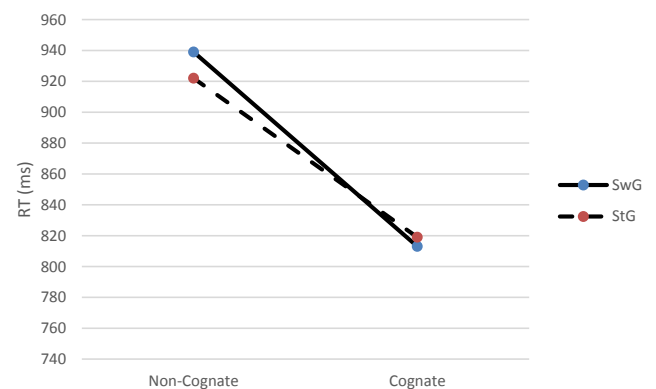


Fig. 2. Naming latencies for cognates and non-cognates for both language varieties in Experiment 1 (Swiss German and Standard German).

cognate effects were not an artefact of frequency effects (see Appendix C). There was no main language effect [$F(1, 4373) = 0.94, p = .331$]. No other interaction was significant ($F_s < 1$).

Discussion

The goal of Experiment 1 was to study switch costs and cognate effects involved in switching between two varieties of a language in highly proficient diglossic speakers who use both varieties aptly, but predominantly in different contexts. The results from diglossic speakers of Swiss German (SwG) and Standard German (StG) revealed that (a) switching was associated with costs, (b) the switch costs were non-paradoxical asymmetrical, i.e. larger for the orally less dominant standard variety (StG), (c) there was no main language-variety effect, (d) there was cognate facilitation for both types of trials in both language varieties, and (e) the cognate facilitation was greater for SwG than for StG.

The magnitude of the switch costs found (SwG: 45 ms, StG: 66 ms) is comparable to bilingual switch costs, as observed in other studies. For example, Costa and Santesteban (2004, Experiments 1, 3 and 4) found switch costs between 35 ms and 81 ms for unbalanced bilinguals (L1 = Spanish: 71 ms, L2 = Catalan: 35 ms; L1 = Korean: 81 ms, L2 = Spanish: 42 ms), and switch costs between 47 ms and 48 ms for highly proficient bilinguals (L1 = Spanish: 47 ms, L2 = Catalan: 47 ms; L1 = Spanish: 47 ms, L3 = English: 48 ms). This is evidence that switching between two varieties in diglossic speakers is as demanding as switching between two languages in bilinguals, suggesting that it requires a similar amount of control.

A similar effect was found for switching between two rather polar language varieties (forming the poles of a continuum) in a study that appeared after our submission of the present report (Kirk, Kempe, Scott-

Brown, Philipp, & Declerck, 2018). The overall amount of switch cost was 68 ms for switching between (Dundonian) Scots and Standard English, and 104 ms for switching between the Aachen dialect (‘Öcher Platt’) and Standard German in older adults. Taken together, the results from their and our two experiments provide convergent evidence that the language control mechanisms in speakers who habitually use a dialect and a standard variety of their language involve the engagement and disengagement of variety control devices, such as language task schemas for the production of words in each of their varieties. Therefore, the lexical entries need to be marked for variety (see also Kirk et al., 2018).

These convergent data also show that the switch-cost effect is generalizable not only to linguistic communities in three different countries (Switzerland, Germany, Scotland), but also to different types of dialect-standard relation. The SwG/StG language use addressed in this study presents a form of *diglossia*, as there are no intermediate forms between standard and dialect (see the discussion above, in the Introduction). In contrast, the Aachen dialect (a Riparian dialect) and Standard German form a standard-dialect continuum with intermediate variants (called a *diaglossic* situation; Auer, 2011), even though nowadays there is a tendency towards dialect loss (Stöckle & Svenstrup, 2011). Similarly, the language use in modern Scottish speech has been described as a ‘bipolar stylistic continuum’ ranging from Scots to (Scottish) Standard English (Aitkin, 1984, cited in Stuart-Smith, 2003). Accordingly, the speakers in these *diaglossic* groups have been described as switching frequently between dialect and standard (Kirk et al., 2018), while the *diglossic* SwG/StG speakers in our experiment use their varieties mainly in different contexts.

The finding of a switch-cost asymmetry in our study, with switch costs being larger when switching from SwG to StG than from StG to SwG, contrasts with the symmetrical switch costs found by Costa and Santesteban (2004) and Costa et al. (2006) for highly proficient bilinguals and by Kirk et al. (2018) for proficient biverbals. The difference in results cannot be explained by proficiency levels. The *diglossic* participants of our study are clearly highly proficient in both varieties and, according to the self-rating data obtained in the four studies, at least as balanced in their proficiency levels as the participants in the other three studies. The difference can also not be due the fact that switching was bilingual in the studies of Costa and colleagues and biverbals in this experiment, as Kirk et al.’s study found symmetric switch costs in biverbals switching. We propose that the difference is due to the fact that Swiss German speakers use SwG and StG predominantly in different contexts, while both dialect users in *diaglossic* environments and speakers of Spanish and Catalan in Catalonia frequently switch in daily life (for Catalonia, see Vila-Pujol, 2007). This proposal is consistent with the idea that speakers that are oriented towards avoiding switching between languages (using one language at a time and switching only when required by a different addressee or context) and therefore practiced at avoiding language conflict develop different control mechanisms than speakers that freely flow back and forth between languages and even code-switch and do not feel the need to avoid switching (Green, 2011). One way to compensate in dual language contexts for an imbalance in baseline activation levels of languages or varieties might be to reduce the accessibility of the stronger language (variety) by lowering its global activation level or increasing the access thresholds, which would be reflected by a global (paradoxical) language effect, as found in Costa and Santesteban (2004) and Costa et al. (2006) studies. In contrast, there was no global language variety effect in this experiment.

The switch-cost asymmetry in this experiment was, however, not paradoxical, favouring switching into the weaker language. Switching into StG involved larger costs than switching into SwG. Even though there was no overall difference in proficiency ratings between the two varieties, SwG was rated slightly stronger in oral expression. While the difference in (oral) proficiency in both varieties is small, there is also an important sociolinguistic difference between them. Whereas SwG is

used in relaxed situations, felt emotionally close and related to one’s identity and is never corrected for errors at school, StG is predominantly used as an instructional medium in class (in addition to writing and inter-cultural communication) and is also a school subject with numerous corrections and evaluations. Unless there is a specific reason to use StG, SwG speakers will always use SwG in conversation. All but two of our participants indicated they generally prefer speaking SwG to StG.

The finding of a non-paradoxical switch-cost asymmetry is puzzling, as at first glance ‘paradoxical’ switch costs are a robust finding for (pairs of) tasks that differ in relative strength – not only in cued language switching, but in a variety of task-switching situations (for reviews, see Kiesel et al., 2010; Umemoto & Holroyd, 2015). There are, however, exceptions to this relation. The switch-cost asymmetry may be reversed if (a) the degree of interference between the two tasks is reduced by separating the onset of stimulus attributes or the response sets for the two tasks (e.g., Yeung & Monsell, 2003), or (b) the stronger task is associated with a higher value resulting from continuous rewards directed at that particular task (Umemoto & Holroyd, 2015). The former factor can be excluded for the *diglossic* switching we are concerned with, as one single picture triggers lexical activation in both varieties and the same response modality and category are used in both cases. The latter factor, however, might play a role. The sociolinguistic difference between SwG and StG speech conforms well to the idea that the SwG speech production is associated with high value, based on frequent reinforcement, and the StG production is not. Importantly, the way how rewards and value are supposed to influence switch-cost asymmetries is via affecting top down control over task performance. This account is in line with the proposal put forward above, in the Introduction, that StG may be more strongly suppressed, because speakers ‘try harder’ to avoid their interference. This proposal is supported by results from structural priming with StG and SwG suggesting that StG structures can be inhibited under certain circumstances (Lüthi et al., 2019a). In addition to the value associated with SwG (in contrast to the sociolinguistic distance, i.e., emotional detachment from StG), the *diglossia*-related endeavour to keep both varieties apart might contribute to why speakers apply an additional amount of top-down control to prevent influences from StG when producing SwG namings.

The observed asymmetry in cognate facilitation (SwG: 127 ms, StG: 102 ms) is consistent with Christoffels et al. (2007) finding in unbalanced bilinguals that cognate facilitation was greater in L1 (84 ms) than in L2 (53 ms), given that SwG is orally the more dominant variety. Accordingly, in contrast to the non-paradoxical switch-cost asymmetry found in our study, the cognate-gain asymmetry is paradoxical – as typical for asymmetrical patterns in switching conditions, in which the baseline asymmetries (found in separate language use; for cognate effects, see Costa et al., 2000) are often reversed (cf. Christoffels et al., 2007, and the discussion above). This is confirmed by the cognate facilitation and cognate-gain asymmetry found in the error analysis, with a more pronounced difference between cognate and non-cognate trials in SwG than in StG. The cognate advantage in both response time and error rate indicates that the non-response language variety is activated along with the response variety. Our finding of a larger cognate effect for SwG than StG is consistent with the argument put forward for bilingualism that the more dominant language (variety) is more open to influences of the other language (variety) (cf. Christoffels et al., 2007; Kroll, Bobb, & Wodniecka, 2006).

This experiment was designed to examine switch costs and cognate facilitation in *diglossic* speakers of Swiss German and Standard German. *Diglossic* speakers showed switch costs with a magnitude comparable to bilingual speakers in previous studies, an asymmetric switch-cost pattern that is different from earlier studies with highly proficient bilinguals, and a cognate-gain asymmetry as found for bilinguals (reversed relative to the switch-cost asymmetry). To be able to compare the findings with a more directly comparable bilingual group, we conducted Experiment 2 with bilinguals of German (*diglossic* SwG/

StG) and Tamil who grew up in German-speaking Switzerland. Also, a particular aim of the experiment was to explore the locus of the cognate facilitation effect in diglossic speakers. If it stems (at least in part) from shared lemmas across varieties, “silent” (SwG/StG) cognates should, again, have a facilitatory effect for both languages in switching between Tamil and (Swiss) German. If the effect is purely phonological, there should be a facilitation for SwG responses, but for Tam responses an inhibition (if processing is cascaded) or no effect at all (if processing is serial).

Experiment 2

Method

Participants

Seventeen native speakers of Bernese German and Tamil living in the canton of Bern participated. Their mean age was 21.1 years (ranging from 18 to 24). All but one had grown up in the canton of Bern (one in Switzerland, no further information given). They all spoke Tamil at home with their parents. They indicated talking to their siblings in SwG ($n = 8$), SwG&Tam ($n = 4$), Tam ($n = 2$), German ($n = 2$, variety not given), or German & some Tam (variety of G not given, $n = 1$). All but one had taken the Tamil test conducted by the Tamil Education Swiss System (TESS). In addition to SwG, StG and Tam, participants knew one to four (mostly two) further languages. All of them reported knowledge of English, all but one of French, and three of Italian. Other languages were mentioned by single persons. Participants' level of language proficiency was assessed for all languages and varieties using the same self-rating proficiency questionnaire as in Experiment 1. The mean scores for Swiss German, Standard German, and Tamil can be found in Table 3. The proficiency profiles between Swiss German and Standard German are again remarkably similar and show a very high proficiency in both language varieties. Only in writing, participants rate themselves slightly, but significantly [$t(16) = 2.22, p = .041$], better in SwG (6.7) than in StG (6.2). In Tamil, participants rate themselves as weaker in general proficiency (5.5) compared to both German varieties (SwG: 6.6, StG: 6.4), and in particular in writing (Tam: 4.5). In the oral modality, the proficiency scores are high in all three varieties (SwG: 6.7, StG: 6.5, Tam: 6.2). However, Tamil is slightly weaker in oral expression, even though the difference to Swiss German only approaches statistical significance [$t(16) = 2.04, p = .058$]. When asked about their preference between the three varieties, 12 participants indicated Swiss German and seven Tamil (including an overlap of two, who explained that their preference depended on the interlocutors). The first and second other languages (ordered by general proficiency or, if that was identical, by mean domain proficiency) had mean general proficiency levels of 5.4 and 3.8, and mean oral expression ratings of 5.3 and 3.4.

Materials

The same pictures as in Experiment 1 were used. No picture had cognate names in Swiss German and Tamil.

The Tamil equivalents of (SwG/StG) cognates and non-cognates were similar in frequency [mean: cognate = 20.3 per million, non-

cognate = 36.5 per million; $t(22) = -0.9, p = .355$] (Sketch Engine corpus, see Kilgarriff et al., 2014; search for bare noun stems). For SwG, frequency norms are not available. SwG cognate words (i.e., words with StG cognates) and their Tamil equivalents tended to be shorter than non-cognates and their equivalents, in terms of syllables [$F(1, 44) = 6.0, p = .019$], and Tamil words were longer than SwG words [$F(1, 44) = 4.1, p = .048$]. However, word length does not affect speech onset latencies when producing single words (Damian et al., 2010). There was no indication of an interaction of variety and cognate status ([mean: cognate_{SwG} = 1.4, non-cognate_{SwG} = 1.9, cognate_{Tam} = 1.8, non-cognate_{Tam} = 2.3; $F < 1$]).

Design and procedure

The design and procedure were identical to those in Experiment 1, with the exception that participants were asked to name the pictures in either Swiss German or Tamil.

Data analysis

Data analysis followed the same principles as in Experiment 1.

Results

Response analysis

Following the criteria in Experiment 1, 16.3% of the trials were excluded as errors and 0.6% as individually correct, but inconsistent responses. Most errors were of the types ‘other language variety’ (9.8%), ‘dysfluency’ (2.9%), or ‘different name’ (e.g., SwG *Haas* ‘hare’ instead of *Chüngu* ‘rabbit’) (2.2%). In addition to swaps between both languages involved, the category ‘other language’ included also Standard German answers in the Swiss German condition (e.g., StG *Kamm* instead SwG *Sträu* ‘comb’), and English answers in the Tamil condition (e.g., *toffee* instead of Tam *inippu* ‘candy’). There was also a considerable number (4.4%) of SwG/StG interference responses, with StG lexemes being phonetically adapted to SwG (e.g., the StG word *Abfall* ‘waste’ produced as *Abfau*, with Bernese *l*-vocalisation).

Trials with recording failures (1.4%) were excluded from further analyses. An error analysis for word errors (‘other word’, ‘other variety’, or ‘article before noun’) using a GLMM with Language, Switch, and Cognate as fixed factors and Participant and Picture as crossed random factors (intercepts and slopes) and binary logistic regression as link function revealed a significant main effect of Cognate [$F(1, 4820) = 13.47, p < .001$], and significant interactions of Cognate and Language [$F(1, 4820) = 104.70, p < .001$], and of Switch and Language [$F(1, 4820) = 6.82, p < .009$]. Fewer errors occurred for cognate trials ($\hat{\mu} = 4.2\%$) compared to non-cognate trials ($\hat{\mu} = 12.8\%$). However, this difference can totally be traced to SwG trials ($\hat{\mu}_c = 2.6\%$ vs. $\hat{\mu}_{non-c} = 23.2\%$); there is no difference between Tam translation equivalents of (SwG/StG) cognates and non-cognates ($\hat{\mu}_c = 6.6\%$ vs. $\hat{\mu}_{non-c} = 6.6\%$). Within Swiss German, there were also more errors in switch than in non-switch trials ($\hat{\mu}_s = 10.2\%$ vs. $\hat{\mu}_{non-s} = 6.6\%$), but not so within Tamil ($\hat{\mu}_s = 6.2\%$ vs. $\hat{\mu}_{non-s} = 7.1\%$). No other main effect or interaction was significant.

RT analysis

Errors and inconsistent correct responses (see above) as well as outliers (4.2%) were excluded from the response time analysis. The effects of Language Variety, Switching, and Cognate Status onto response were analysed using a GLMM with Gamma distribution and Log link, with Participant and Picture as crossed random factors (intercepts and slopes), implemented using the GENLINMIXED procedure in SPSS 24.0.0.1 (estimation with linearisation-based methods). Table 4 lists the mean response latencies as a function of fixed-factor combinations.

There were significant main effects of Switching [$F(1, 3855) = 94.69, p < .001$] and Cognate Status [$F(1, 3855) = 32.82, p < .001$], indicating that switch trials were slower than non-switch trials (see Fig. 3), and that cognates were

Table 3
Proficiency questionnaire (SwG, StG, & Tam) in Experiment 2.

	Swiss German		Standard German		Tamil	
	Mean	SD	Mean	SD	Mean	SD
Comprehension	6.6	0.6	6.8	0.5	6.4	0.9
Oral expression	6.7	0.6	6.5	0.9	6.2	1.1
Reading	6.5	0.8	6.6	1.1	4.8	1.6
Writing	6.7	0.5	6.2	0.9	4.5	1.9
General proficiency	6.6	0.5	6.4	0.6	5.5	1.1

Table 4
Estimates for Mean Response Latencies in ms (with Standard Errors) [and 95% Confidence Intervals] as a Function of Language Variety, Switching, and Cognate Status for Experiment 2.

	Swiss German		Tamil	
	Non-Cognate	Cognate	Non-Cognate	Cognate
Non-switch	965 (222) [615, 1513]	866 (199) [552, 1359]	980 (225) [625, 1538]	893 (205) [569, 1401]
Switch	1049 (241) [668, 1646]	979 (225) [624, 1537]	1052 (242) [671, 1651]	954 (219) [608, 1497]

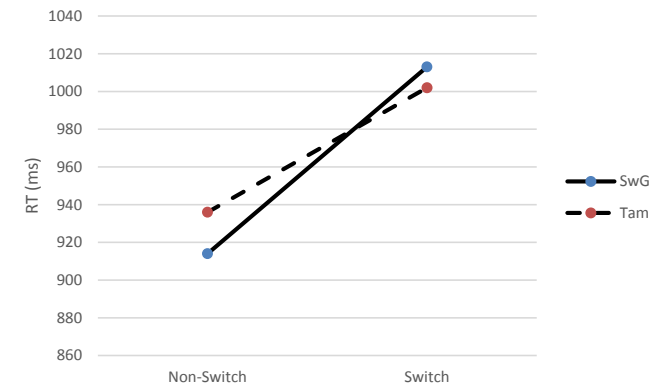


Fig. 3. Naming latencies for switch and non-switch trials for both language varieties in Experiment 2 (Swiss German and Tamil).

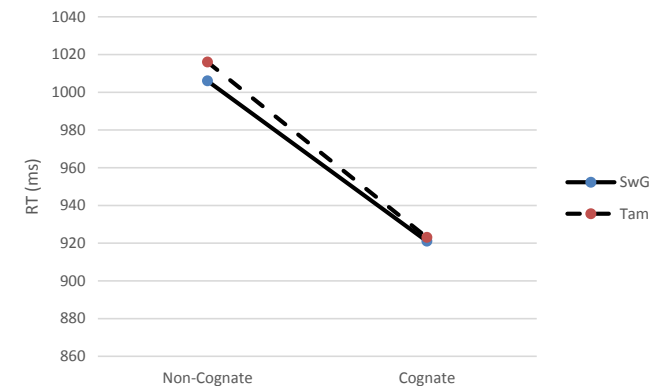


Fig. 4. Naming latencies for (SwG/StG) cognates and non-cognates, and their translation equivalents, for both language varieties in Experiment 2 (Swiss German and Tamil).

produced faster than non-cognates (see Fig. 4). There was no main language effect [$F(1, 3855) = 0.45, p = .504$]. The Switching effect was qualified by an interaction with Language [$F(1, 4355) = 3.90, p = .048$], showing slightly asymmetrical switch costs, with $\text{Tam} \rightarrow \text{SwG} > \text{SwG} \rightarrow \text{Tam}$ (see Fig. 3). Simple contrasts confirm for both language varieties clear, statistically reliable cognate gains [SwG: 85 ms ($SE = 26$ ms), $t(3855) = -3.20, p = .001$; Tam: 93 ms ($SE = 28$ ms), $t(3855) = -3.37, p = .001$] and switch costs [SwG: 99 ms ($SE = 26$ ms), $t(3855) = -3.82, p < .001$; Tam: 66 ms ($SE = 19$ ms), $t(3855) = -3.44, p = .001$], with switch costs differing only in size. The cognate facilitation effect was significant for all type-of-trial and language combinations [no-switch/SwG: 98 ms ($SE = 29$ ms), $t(3855) = -3.40, p = .001$; no-switch/Tam: 87 ms ($SE = 27$ ms), $t(3855) = -3.23, p = .001$; switch/SwG: 69 ms ($SE = 30$ ms), $t(3855) = -2.30, p = .022$; switch/Tam: 99 ms ($SE = 33$ ms), $t(3855) = -2.99, p = .003$]. No other interaction was significant ($F_s < 1$).

Discussion

The goal of Experiment 2 was to investigate switch costs and cognate effects involved in switching between one of the two varieties studied in Experiment 1, and another language in highly proficient bilingual speakers who also use the language varieties predominantly in different contexts. The results from bilingual speakers of Swiss German (SwG) and Tamil (Tam) revealed that (a) switching was associated with costs, (b) the switch costs were asymmetrical, with the more dominant language variety (SwG) showing the larger costs, (c) there was no main language effect, (d) there was cognate facilitation (from SwG/StG cognates) for both types of trials in both languages, and (e) the cognate facilitation was of comparable magnitude for SwG and Tam.

Switch costs were asymmetrical; however, the switch-cost asymmetry was in the opposite direction from Experiment 1: Switching into SwG, the more dominant language (variety) was harder, suggesting that it was suppressed more strongly than Tam. Again, as in Experiment 1, there was no main language effect. Taken together, the bilingual switching pattern corresponds to the classic “paradoxical” switch cost pattern (observed in most studies with unbalanced bilinguals), i.e. higher costs for switching into the dominant language plus no general language effect. As the bilinguals in our study are highly proficient in both languages, the results support the view that lexical selection may compete between lexical entries from both languages (be language-unspecific) even in highly proficient bilinguals. This finding contrasts with proposals that either high bilingual proficiency (Costa & Santesteban, 2004; Costa et al., 2006) or an early onset of bilingualism (Martin et al., 2013) leads to a particular mechanism underlying switching and language control, different from that in other bilinguals. The results rather support the idea put forward above (Introduction) that the absence of asymmetrical switch costs associated with a paradoxical language effect (disadvantage for the more dominant language) in some groups of bilinguals might be the consequence of a high level of switching experience. This would also explain why asymmetrical switch costs may be observed in unbalanced/late bilinguals (e.g., Christoffels et al., 2007).

The view that language control in deliberate language switching is based on an inhibition mechanism suppressing the nonresponse language in our group of bilinguals is also supported by the cognate-effect data. The fact that there was cognate facilitation even for Tam responses suggests that StG was activated along with the two varieties (SwG and Tam) used for picture naming, and that competition from two cognate words is easier to suppress than competition from two words that are not cognates unto themselves. The cognate gain in Tam responses (92 ms) was not smaller than in SwG responses (85 ms).

The finding of a SwG/StG cognate facilitation for Tam responses suggests that SwG and StG have shared lexical representations and that the cognate facilitation stems from having fewer lexical representations to suppress. If the cognate facilitation in SwG word production resulted from an activation of the corresponding phonological segments in StG, we would expect a *negative* cognate effect in Tam responses, because the alternative (SwG) name’s phonological form would receive additional activation from StG.

The conclusions drawn from the response time analysis are also consistent with the results from error analysis. The finding that fewer errors occurred for cognate trials than for non-cognate trials in SwG responses (but not in Tam responses) is compatible with a lexical architecture that includes integrated representations of SwG/StG cognates, because there would be fewer lexical entries among which to select for production. The fact that there were more errors in switch than in non-switch trials for SwG is consistent with the view that the change to a new word-production schema (according to the IC model; Green, 1998) is associated with an activation of previously suppressed lexical items. In non-switch trials, however, the system remains attuned to the language variety in use. For bilingual plus diglossic speakers, switching from one language to one of the two varieties of the other

language is so demanding that not only higher response times but also more errors occur.

General discussion

In the present study, we investigated language switching in diglossic speakers, that is, speakers of two rather sharply distinguished varieties of a language without intermediate forms, in order to explore the mechanisms involved in lexical access and language control in diglossic speech production. In Experiment 1, speakers of Swiss German (SwG) and Standard German (StG) named pictures, switching between SwG and StG as indicated by a colour cue. Experiment 1 addressed switch costs and cognate gains, variety effects, and their interactions. In Experiment 2, bilingual speakers of Tamil (Tam), SwG and StG performed the switching task in SwG and Tam. Experiment 2 both provided a direct comparison with bilingual switching (between one of the two varieties in question and another language) and allowed us to assess the cognate effects of the variety that was not overtly used.

Switch costs in bivarietal and bilingual switching

Switching between two diglossic varieties of a language results in costs of a similar magnitude as switching between two different languages. This finding is consistent with the assumption that diglossic speakers have to control for selecting words from the lexicon of the intended variety and not the other variety, as bilingual speakers do in restricting lexical choice to lexical entries from one language. Taken together, the convergent results from Kirk et al. (2018) study on switching between the polar varieties of a language-dialect continuum (Aachen dialect & Standard German, and Dundonian Scots & Standard English) and ours on switching between diglossic varieties (SwG & StG) are compatible with the idea that switching between two rather distinct varieties of the same language may work in the same way as switching between different languages. An open question, inviting further research, is whether the same type of control applies in switching between language varieties in the form of smooth transitions, such as moving back and forth along probabilistic distributions of variants, such as in style-shifting.

Language control, according to the IC model (Green, 1998), is achieved via language schemas and inhibition of the non-selected language (variety), with switch costs reflecting the time needed for re-setting the cognitive system and activating a different language task schema plus overcoming the previous inhibition. As inhibition for lexical selection operates on lexical representations associated with language tags, an extension of the model to varietal language control, as suggested in this paper, implies that (other than assumed in current monolingual models, e.g., Levelt et al., 1999) one concept may be linked with multiple lexical representations in a language, associated with language tags signalling variety, similarly proposed for the translation-equivalent lexical representations in the two languages of a bilingual (see Costa, 2005).

Paradoxical and non-paradoxical switching patterns

Two switching patterns have been found in research on bilingual switching, and both involve “paradoxical” effects, in the sense of reversing the baseline response time discrepancies found in single language contexts (e.g., blocked conditions in an experiment). In contrast to the faster performance in the stronger language found there (e.g., Christoffels et al., 2007), in switching conditions, paradoxical global language effects (slower picture naming in the stronger language) or paradoxical switch costs (larger switch costs for switching into the stronger language) may occur, resulting in two switching patterns: (1) paradoxical language effect only (with symmetrical switch costs) and (2) paradoxical asymmetrical switch costs (with or without additional reversed language effect) (see introduction for a review). Symmetrical

switch costs have been argued to be indicative of a different lexical selection mechanism, which does not require inhibitory control, for highly proficient bilingual (Costa & Santesteban, 2004; Costa et al., 2006), or early bilinguals (Martin et al., 2013).

In contrast to this proposal, we obtained the “classical” switch-cost asymmetry, typically found for unbalanced bilinguals (e.g., Costa & Santesteban, 2004; Meuter & Allport, 1999), even in early-onset, highly proficient bilinguals. This finding provides evidence in support of the view that even in early-onset, highly proficient bilinguals lexical candidates from both languages compete for selection (see also Kroll, Bobb, Misra, & Guo, 2008). As the Tamil/German bilinguals in our study use their languages (and varieties) predominantly in different context, this finding is rather consistent with the idea advanced in the introduction that the switching pattern ‘paradoxical language effect plus symmetrical switch costs’ is related to frequent switching behaviour. This was based on the reverse observation that the symmetrical switch costs can also be found in unbalanced bilinguals who switch frequently between their languages (Christoffels et al., 2007).

Accordingly, we propose that the mechanisms underlying the switching performance in bilinguals depend on the way in which they typically use their two languages: whether involving a large amount of switching or avoiding switching. According to this hypothesis, bilinguals who use their languages frequently in dual-language contexts may come to develop a global biasing mechanism favouring the otherwise disadvantaged language (the L2, or the less dominant language), as reflected in a paradoxical language effect. Such a biasing mechanism, which might arise at the level of the lexical system (in terms of reduced accessibility; cf. Costa & Santesteban, 2004) or at the level of the language task schema (in terms of being more readily activated due to top-down control; cf. Green, 1998), would reduce the otherwise expected imbalance in reactive inhibition between languages, leading to symmetrical switch costs. In contrast, bilinguals who use their languages frequently in single-language contexts will not develop such a mechanism. Accordingly, their switching performance will follow the second switching pattern (paradoxical switch-cost asymmetry plus no paradoxical language effect), as in the Tamil/German speakers in Experiment 2.

The idea that differences in switching patterns arise from switching experience rather than from proficiency level or onset age is consistent with the adaptive control hypothesis (Green & Abutalebi, 2013; see also Green, 2011). This states that recurrent patterns of linguistic interaction lead to a persistent adaptation of the language control processes to the demands placed on them by those contexts, such as interference control in single-language contexts or task disengagement in dual-language contexts. Following this line of thought, we propose that the different switching patterns can be explained by the involvement of different mechanisms for mastering the switching task – without the need to postulate two utterly different mechanisms of lexical selection itself in different groups of bilinguals, as proposed by Costa and Santesteban (2004). Instead, we propose that lexical selection is always language-unspecific in bilingual production and requires control mechanisms for ensuring the restriction of lexical output to one language, as discussed further below (section on loci of control).

In contrast to the bilingual switching in Experiment 2, the diglossic switching in Experiment 1 followed a “non-paradoxical” pattern, in that the orally more dominant language variety, SwG, was neither subject to a paradoxical main language effect nor to paradoxically higher switch costs. There was a switch-cost asymmetry, yet it was *in favour* of SwG, i.e., switching into SwG showed the smaller costs. We must take into account that “non-paradoxical” switch-cost asymmetries (other than suggested by the notion) are the exception, and not the default case, in studies about switching between two tasks of differing strengths, and also that there is wide agreement in the task-switching literature that switch-cost asymmetries stem from persisting transient effects of the previous task in interaction with task strengths and current control processes (e.g. Kiesel et al., 2010; Monsell, 2003; Yeung & Monsell,

2003). Therefore, an account of the found pattern needs to consider an interplay of the relevant factors and explain why the usual relation is reversed. Even though the aftereffects of a previous task could in principle consist of persisting activation of the previous task or persisting inhibition of the previously suppressed task, there is separate evidence for inhibitory control processes in general (for a review, see Kiesel et al., 2010) and in bilingual speech in particular (Bobb & Wodniecka, 2013; Kroll et al., 2008). The factors taken into account in the following explanation that we offer are accordingly: (1) persisting inhibition from the previous trial, (2) language (variety) strengths, and (3) top-down cognitive control.

There is no indication of a strength set-off in terms of a global activation increase of the (orally slightly) weaker variety, StG; i.e., there is no global speed effect. As indicated by the non-paradoxical switch-cost asymmetry, the inhibition of StG as non-selected variety (i.e., in SwG trials) – to be overcome in switching – is stronger than the inhibition of SwG as non-selected variety (i.e., in StG trials). We assume that this is the case because of a greater application of top-down cognitive control to suppress StG during SwG trials. The reason for this is supposed to be the higher value associated with speaking SwG, i.e., the sociolinguistic distance (emotional detachment) from StG, in addition to the idea to keep varieties separate, which has a strong presence in society. Participants seem to put more weight on avoiding the intrusion of StG lexical items in the SwG condition than vice versa, in accordance with the largely accepted idea to ban StG words from SwG dialogue and the wide-spread preference of SwG for oral communication, which is also present in our participants.

More generally, this proposal attempts to specify a possible role for top-down control in regulating language performance. While it can be assumed that speakers usually apply the minimum amount of top-down control required to perform the language task with a sufficient accuracy, as applying top-down control is effortful (cf. Yeung & Monsell, 2003), there may be particular circumstances and conditions where enhanced top-down control is exerted. We suggest that one of those, in terms of theories of action selection based on hierarchical reinforcement learning (see Umemoto & Holroyd, 2015), is a higher reinforcement value of one of the linguistic options (such as being liked more). Importantly, the level of linguistic options addressed here is not the level of individual linguistic variants, but a higher level, such as ‘talking in SwG’, corresponding to language task schemas in Green (1998) IC model.

Cognate effects and the representation of cognates in diglossic speakers

The cognate facilitation in diglossic speakers switching between SwG and StG indicates more directly that lexical items from both varieties are activated in parallel. The finding that cognate facilitation was larger for SwG than for StG is consistent with Christoffels et al. (2007) observation of a larger cognate facilitation effect in L1 than in L2 in bilingual switching. The conclusion that the stronger language is to a higher degree subject to the other, also active, language’s influences than vice versa (see Kroll et al., 2006), therefore extends to language varieties in the diglossic case.

The finding of a SwG/StG cognate facilitation even in bilingual switching between Tam and SwG suggests that both German varieties are activated, even when only one of them is overtly used across all trials and, accordingly, the SwG and the Tam task schemas would suffice to be preactivated.

This SwG/StG cognate facilitation in Tam/SwG switching occurred not only in SwG, but also in Tam trials, and the magnitude of the cognate gains was similar in both. This is supportive of the notion that cognates between two varieties of a language may have integrated representations at the lexical level, consistent with evidence obtained for SwG and StG with the syntactic-priming paradigm by Lüthi et al. (2019b). If the cognate effect arose at the phonological level, we should expect an inhibitory effect of cognate SwG/StG translation-

equivalents for Tam responses in the case of cascaded processing, as the phonological form of the alternative (SwG) name would receive additional activation from the (also activated) StG lexical representation. In the case of serial, discrete processing, there should be no “silent” (SwG/StG) cognate effect on Tam responses at all, as only the selected (Tam) lexical node would send activation to the phonological level. The idea that SwG and StG have integrated lexical representations is also supported by the error patterns in both experiments. Both the SwG/StG and the Tam/SwG/StG speakers produce more errors in non-cognate than in cognate trials, but the Tam/SwG/StG speakers only do so in SwG (not in Tam) responses, and the SwG/StG speakers more so in SwG than in StG responses. Both the cognate RT advantages and the cognate error advantages can be accounted for by the assumption of shared lemma representations. A higher number of competing lexical representations takes more time to suppress and gives more opportunities for erroneous selection.

The fact that the Tam/SwG/StG speakers produced far more errors in SwG non-cognate trials (23.2%) than the SwG/StG speakers (5.1%) indicates that they have a less clear separation of the two German varieties in their lexical representation, with more lexical items being represented as cognates, suitable for use in both varieties (in a morpho-phonologically adapted form).

Loci of language control

Current accounts of lexical access in bilingual speakers favour the assumption that different mechanisms of language control work together (see, e.g., Christoffels et al., 2007) and that there are multiple levels of control (Green, 1998). Taken together, the results from our study on switching between diglossic varieties and Kirk et al.’s (2018) study on switching between polar varieties suggest that, at least, rather distinct varieties of a language are also governed by multiple mechanisms and multiple levels of control.

The pattern of switch costs and cognate gains observed is compatible with the assumption that the candidates in the non-target variety are inhibited and this inhibition carries over to the next trial (proactive inhibition), if we assume that the amount of inhibition applied is not just a passive reflection of the strengths of lexical links but can be actively modulated by top-down cognitive control. The idea that inhibitory control is involved is supported by the finding of “silent” cognate effects in bilingual production (Experiment 2), which suggests that SwG/StG cognates share lexical representations across varieties. Accordingly, the cognate effects in both experiments can be explained in terms of less competition from alternative, translation-equivalent names for lexical selection. In the case of Tam naming in Experiment 2, just one, and not two, German naming alternatives have to be rejected for lexical selection. A control mechanism in terms of variety-specific selection would face difficulties to account for the cognate effect in Experiment 2. The control mechanism restricting lexical access to the intended variety needs to be able to deal with the activation of competing naming alternatives.

Even though the results are compatible with an account in terms of inhibitory control, there appear to be limits to the effects of inhibitory control. First, the cognate gains were of a similar magnitude as the switch costs in Experiment 2 and they were even considerably higher than the switch costs in Experiment 1. Second, there was no significant interaction between switch costs and cognate gains. Thus, there was no indication of a reduction of cognate gains in repeat (non-switch) trials. In Christoffels et al. (2007) study on bilingual switching, the cognate effect was even larger in non-switch trials. Clearly, the inhibition of the alternative lexical entries, carrying over from trial to trial, does not seem to be strong enough to prevent influences from competing items onto lexical selection.

The second locus of language control, as already mentioned, is top-down control. As discussed above, one possible reason to apply more than the minimum control bias required might be a positive value

associated with one of the varieties, such as the sociolinguistic significance that SwG enjoys in Switzerland (at least for oral communication). More generally, this explanation is in line with [Yeung and Monsell \(2003\)](#) model of task switching, which addresses the question of why the repetition benefit for the weaker task is not as large as the switch cost for returning to the stronger task in (paradoxical) asymmetric situations and assumes that it is offset by less top-down control in non-switch trials. If, for some reason, more top-down control is exerted in non-switch trials, this should lead to a reduction or even a reversal of the switch-cost asymmetry.

A third mechanism of control is a global biasing of the system towards a certain language (variety) by a global increase of activation (or lowering of thresholds) of the items of a weaker language (e.g. [Costa et al., 2006](#)), proactive inhibition ([Christoffels et al., 2007](#)), or a pre-activation (readying of advancement) of the language schema (which would still await input for the activation of lexical representations, cf. [Green, 1998](#)). Our result pattern is supportive of the proposal put forward by [Christoffels et al. \(2007\)](#) that such a mechanism of sustained language control might depend on the amount of switching experience in life. The fact that we did not find a paradoxical global language effect, but a paradoxical switch-cost asymmetry in bilingual speakers who use their language mainly in different contexts suggests that inhibitory control can occur even in early-onset and highly proficient bilinguals

and the control mechanism of global biasing may depend on the amount of regular switching experience.

Taken together, the available evidence (on switch costs, cognate gains, and global effects in language and variety switching) suggests an interplay of multiple levels and mechanisms of control in restricting lexical selection to one language or one language variety: activation of lexical representations, top-down control, activation and suppression of language schemas, global biasing (of language schema or lexical systems), and inhibition of lexical representations. The interplay of mechanisms may depend on the strengths of the languages or varieties involved, the life-long experience in dual vs. single language (variety) contexts, the sociolinguistic relation of languages or varieties and the current language-use situation.

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Appendix A: Description of the linguistic situation in German-speaking Switzerland

The SwG dialects form part of the Alemannic dialect continuum, which stretches over parts of France, Switzerland, Germany, Italy, and Austria, and Liechtenstein, and is split into Low, High, and Highest Alemannic. It is phonologically characterised by particular vowels preserved from Middle High German, such as the diphthongs /iə/, /uə/, /yə/ where StG has monophthongs (/i:/, /u:/, /y:/), and the long monophthongs (/i:/, /u:/, /y:/) where StG has diphthongs (/ai/, /au/, /ɔy/; e.g., *Schwyz* = *Schweiz* ‘Switzerland’, *Huus* = *Haus* ‘house’, *dütsch* = *deutsch* ‘German’). There are other phonetic/phonological differences, such as long vowels before double consonants and certain consonant clusters where StG has short vowels, which may be more or less pronounced depending on the degree to which pronunciation is influenced by norms from Germany or closer to the dialect usage. In syntax and morphology, differences include the nominative/accusative syncretism in SwG, but not StG, the existence of a preteritum tense in StG, but not SwG, a number of different word-formation morphemes, distinct relative pronouns, differences in inflection forms, and a difference in verb positioning in perfect sentences with modal verbs, among other things. Further linguistic features depend on the particular dialect spoken. In the vocabulary, there are many lexical similarities (cognate words with considerable phonological overlap), but also many differences. In particular, in areas such as human life, household, agriculture, crafts, and nature, there may be a wide diversity among SwG dialects (e.g., StG (die Blumen) giesen ‘to water (the flowers)’ = SwG: *sprütze*, *Wasser gää*, *z trinke gää*, *tränke*, *bschütte*, *netze*, *wässere*, etc.). We are not aware of any estimates of the degree of lexical overlap between SwG and StG.

For oral communication, SwG is generally preferred, while StG is used for most written texts (in textbooks and journals almost exclusively). Swiss German speakers use the dialects in basically all conversations when talking among themselves. Spoken StG is used when required or necessary – mainly in class (except in small groups and interim conversations) or when talking to non-native speakers, or when there is a clear relation to written language, as when quoting from written texts, or when giving a speech prepared in writing. Accordingly, the main reasons for code-alternation are a change in interlocutors (with vs. without knowledge of SwG), function (e.g., a written-language related quotation in StG within a SwG conversation), or an interim private conversation in SwG within a class held in StG; for a review, see [Vorwerg, 2019](#)). Code-switching (switching within a speech event), however, is a rare occurrence in the SwG/StG speech community. Speakers usually rather try to avoid mixing SwG and StG, and the history of linguistic and language-ideological debates in German Switzerland has seen a number of pleas for a sharp distinction between dialect and standard language, indeed (see [Sonderegger, 2003](#)).

Appendix B: Stimuli list

(Swiss/Standard German) Non-cognates				(Swiss/Standard German) Cognates			
Swiss German	Standard German	Tamil	English translation	Swiss German	Standard German	Tamil	English translation
Stäge	Treppe	Padi	‘stairs’	Muu	Mund	Vaai	‘mouth’
Täfel	Bonbon	Inippu	‘candy’	Oug	Auge	Kann	‘eye’
Anke	Butter	Badder	‘butter’	Hang	Hand	Kai	‘hand’
Härdöpfu	Kartoffel	Kilangu	‘potato’	Stueu	Stuhl	Kathira	‘chair’
Ghüder	Abfall	Kuppa	‘waste’	Blueme	Blume	Puu	‘flower’
Sträu	Kamm	Siippu	‘comb’	Schlüssu	Schlüssel	Tirappu	‘key’
Güggu	Hahn	Seeval	‘cock’	Bett	Bett	Kaddil	‘bed’
Lüti	Klingel	Bell	‘bell’	Baue	Ball	Pandu	‘ball’
Ross	Pferd	Kuthirai	‘horse’	Hung	Hund	Naai	‘dog’
Chüngu	Kaninchen	Muyal	‘rabbit’	Chue	Kuh	Maadu	‘cow’
Töff	Motorrad	Motorcycle	‘motorbike’	Tüür	Tür	Kathavu	‘door’
Rüebli	Karotte	Carrot	‘carrot’	Mässer	Messer	Katthi	‘knife’

Appendix C: A Posteriori response time profiles over word frequency

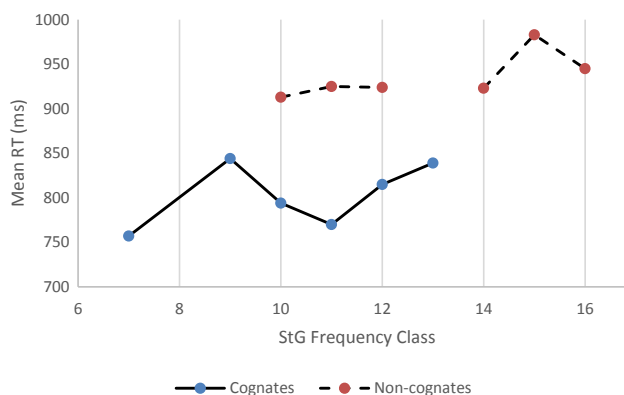


Fig. C.1. Naming-latency profiles over (StG) word frequency for cognates and non-cognates for Swiss German in Experiment 1. The frequency class of a word describes its frequency in relation to the frequency of the most frequent word, with the most frequent words belonging to frequency class 0.

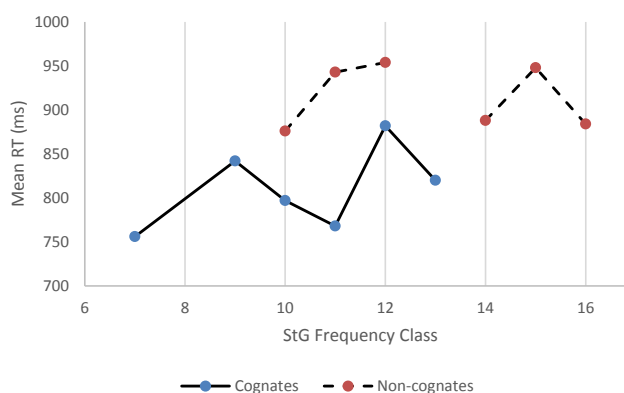


Fig. C.2. Naming-latency profiles over (StG) word frequency for cognates and non-cognates for Standard German in Experiment 1. The frequency class of a word describes its frequency in relation to the frequency of the most frequent word, with the most frequent words belonging to frequency class 0.

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